

IS 300:1968 — National Flag of India (Silk Khadi) Specification

Structured reference sheet: dimensions, materials and construction requirements, summarized for technical/compliance-checking use.
Note: This is a working summary of published requirements, not a reproduction of the full IS 300:1968 standard text. Refer to the original BIS document for complete, authoritative clauses.

Scope

IS 300:1968 (Second Revision) prescribes the design, constructional details and material requirements of the National Flag of India made from hand-reeled, hand-woven silk khadi bunting.

Design

- Shape: rectangular, length:width ratio = 3:2 (other ratios permitted for special purposes)
- Tricolour panel: top = India saffron (Kesari), middle = white with navy-blue Asoka Chakra at centre, bottom = India green
- Large flags may use sub-panels of smaller cloth widths joined by lapped seam, raw edges turned in

Materials — Yarn & Cloth

- Bunting warp yarn: ~5.6 tex (50 denier); weft yarn: ~8.4 tex (75 denier), hand-reeled silk
- Cotton khadi duck (used for sleeve): hand-spun from clean carded cotton, ~38 tex × 3 (Nm 26/3) for warp and weft
- Colours must visually match the sealed standard flag held by the Chief Inspectorate of Textiles & Clothing, Kanpur

Table 1 — Silk Khadi Bunting Particulars

Sl	Characteristic	Requirement	Test Method
i)	Ends/dm	504 ± 8	IS 1963-1961
ii)	Picks/dm	414 ± 16	IS 1963-1961
iii)	Weight, g/m²	68 ± 4	IS 1964-1961
iv)	Breaking load, 50×200mm strips, kg Min (Warp / Weft)	45 / 60	IS 1969-1961

Table 2 — Silk Khadi Bunting Requirements (colour fastness, pH, shrinkage)

Sl	Characteristic	Requirement	Test Method
i)	Colour fastness to light	6 or better	IS 686-1957 or IS 2454-1967
	Colour fastness to washing	4 or better	IS 687-1966
ii)	pH value	6.0 Min, 7.5 Max	Cold method, IS 1390-1961
iii)	Dimensional changes	2.5% Max	IS 3561-1966

Table 3 — Cotton Khadi Duck Requirements

Sl	Characteristic	Requirement	Test Method
i)	Ends/dm	158 ± 4	IS 1963-1961
ii)	Picks/dm	134 ± 12	IS 1963-1961
iii)	Weight, g/m ²	355 ± 35	IS 1964-1961
iv)	Breaking load, 50×200mm strips, kg Min (Warp / Weft)	48 / 43	IS 1969-1961

Sewing Thread

- Heavy silk thread: 3 yarns of pure thrown silk, Z-twisted (yarns S-twisted); colour matches panel; length ≥ 9060 m/kg; breaking load ≥ 3.5 kg on 60cm test length
- Hemp cordage stitching: cotton thread per Variety No. 19 of IS 1720-1960 [27 tex × 3 × 3]

Table 4 — Hemp Cordage Requirements

Nominal Size (Circumference), mm	No. of Strands	Nominal Linear Density, ktex	Breaking Load /1m, kg Min
19	3	25	150
25	3	52	300
32	3	64	355
38	3	105	590

Wooden Toggles

Made from well-seasoned timber. Recommended species: Haldu, Padauk, Shisham, Sal, Teak.

Table 5 — Flag & Wooden Toggle Dimensions (all mm; tolerances ±1.5-2%)

Flag Size No.	Leng th (A)	Widt h (B)	Hem Dept h (C)	Slee ve D	Slee ve E	Cor d L	Cor d M	Cor d N	Togg le P	Togg le Q	Togg le R	Togg le S	Togg le T
1	6300	4200	25	65	40	38	65	140	125	30	22	15	14
2	3600	2400	15	50	40	32	65	130	105	25	17	7	12
3	2700	1800	15	45	40	25	65	120	90	20	13	5	10
4	1800	1200	15	40	40	19	65	110	75	17	10	4	8
5	1350	900	15	40	40	19	65	110	65	17	10	4	8
6	900	600	15	40	40	19	65	110	65	17	10	4	8

Table 6 — Asoka Chakra Dimensions (all mm; tolerance $\pm 2\%$ or $\pm 0.5\text{mm}$)

Flag Size No.	J (mm)	K (mm)	L (mm)	N (nominal spokes)
1	1295	1120	112	14
2	740	640	64	8
3	555	480	48	6
4	370	320	32	4
5	280	242	24	3
6	185	160	16	2

Chakra design: 24 equally spaced spokes, navy blue, embroidery thread per Variety No. 7 of IS 1803-1961, screen printed/printed/stencilled/embroidered, fully visible on both sides.

Construction Requirements

- Machine stitching: lock stitches, ≥ 6 stitches per centimetre
- Panels joined with $\pm 2\text{mm}$ tolerance on 10mm lapped seam depth
- Raw edges on 3 sides turned twice and hemmed; sleeve attached on 4th side
- Four corners reinforced with triangular bunting pieces of matching construction/colour
- Hemp cordage inserted in sleeve, hand-stitched at top/middle/bottom, with spliced loops at both ends (toggle loop at top)
- Toggle spliced into cordage at top before sewing into sleeve

Sealed Sample & Marking

All flags must generally conform to the sealed standard flag held by the Chief Inspectorate of Textiles & Clothing, Kanpur.

Flags marked on sleeve with: flag size no., length $\times$ width (mm), manufacturer name/trademark, year of manufacture, other buyer-required info. ISI Certification Mark use optional, governed separately.

Table 7 — Sampling: Flags to be Selected & Permissible Non-Conforming

Number of Flags in Lot	Number to be Selected	Permissible Non-conforming
Up to 25	3	0
26-50	5	0
51-100	8	0
101-150	13	0
151-300	20	1
301-500	32	1
501-1000	50	2
1001-3000	80	3
3001 and above	125	5

Criteria for Conformity

- For dimensional/construction requirements (9.4): observed failures must not exceed Table 7 col 3 limit
- For material-level requirements (9.5 — bunting, duck, thread, cordage): conformity criteria as agreed between buyer and manufacturer

Key Reference Standards Cited

IS 1963-1961 — Ends/picks per unit length in woven fabrics

IS 1964-1961 — Weight per square/linear metre of fabrics

IS 1969-1961 — Breaking load & elongation at break (constant-rate-of-traverse)

IS 686-1957 / IS 2454-1967 — Colour fastness to daylight / artificial (xenon) light

IS 687-1966 — Colour fastness to washing

IS 1390-1961 — pH value of aqueous extracts of textile materials

IS 3561-1966 — Dimensional changes on washing of woven silk fabrics

IS 1720-1960 — Cotton sewing thread, bleached or dyed

IS 1803-1961 — Cotton embroidery thread, bleached or dyed

IS 196-1966 — Atmospheric conditions for testing

Appendix A — Method for determination of breaking load of hemp cordage (constant-rate-of-traverse test, tension = 3% of specified breaking load, report uses mean – 0.4×range ≥ specified minimum)